Adrenal & Retroperitoneum

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 145	Surgical & Radiographic Anatomy of the Adrenals
Ch 146	Adrenal Disorders: Pathophysiology, Evaluation & Medical Management
Ch 147	Surgery of the Adrenal Glands
Ch 140	Retroperitoneal Tumours
★	Putting it together: adrenal and retroperitoneal decisions

145 Surgical and Radiographic Anatomy of the Adrenals

- P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — *relations, arteries, veins, zones, embryology*

1 · Embryology

- Adrenogonadal ridge: cephalic → cortex, caudal → gonad. Neural crest enters weeks 8–10.
- At birth gland is **2×** adult weight; **fetal zone involutes in year 1**, replaced by ZG/ZF under ACTH. Zona reticularis (ZR) reaches adult volume at 6–8 y.
- **Renal agenesis**: adrenal **discoid**.
- **Adrenal rests**: ≈1% adults (near coeliac axis); along gonadal descent in 7.5–15% neonates (regress). In CAH → **testicular masses**.

2 · Zones and hormones

Cortex ≈ 90%, medulla ≈ 10%. †Outside → in: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% (aldosterone; small clustered cells); fasciculata **~80%** (cortisol; lipid-laden "clear cells"); reticularis 5–7% (DHEAS; anastomosing cords); medulla ~10% (chromaffin cells).
- Medulla = **modified sympathetic ganglion**; chromaffin cells = postganglionic neurones without axons.

3 · Relations and blood supply

- ≈5 g, equal in both sexes. Right gland **retrocaval**, directly above upper pole; left medial to upper pole.
- **Exposure** — right: liver up, IVC medial, duodenum/kidney down. Left: pancreas + spleen up, splenic flexure medial.
- Three arteries → 10–50 capsular twigs: **many in, one vein out**. Cortical blood drains into medulla → cortisol drives **epinephrine synthesis**.
- **Right vein**: short, into **posterior IVC** → avulsion. **Left**: into left renal vein, joined by inferior phrenic.

> **KEY POINT**
> Right side is riskier: high, retrocaval, short vein — little working length; avulsion tears the cava.

5 · Imaging appearance

- **CT**: best spatial resolution; inverted **V/Y**, limbs 3–6 mm (thinner than crus). Unenhanced **≤10 HU** = lipid-rich. Morphology only.
- **MRI**: best contrast; coronal confirms origin. T1 slightly below liver; on **T2 resembles fat**.
- **US**: hypoechoic; role = solid vs cystic.
- **Angiography**: adrenal venous sampling (AVS) = only **functional** study; cavography for intraluminal tumour; arteriography rarely needed.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · Applied anatomy and physiology

- P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — *veins, arteries, zones, PNMT, COMT → metanephrines*
- Anomalous venous drainage in up to **9%** – matters for adrenal vein sampling (AVS). Triple supply → partial adrenalectomy rarely infarcts remnant.
- **Pituitary failure or exogenous steroids spare aldosterone** (glomerulosa obeys angiotensin II/K^+); inner zones atrophy without ACTH.
- Aldosterone raises volume, **not serum sodium**.
- PNMT almost unique to medulla → mainly adrenal tumours make epinephrine. >90% of circulating metanephrine is adrenal.
- α2 suppresses insulin → hyperglycaemia, then post-resection rebound hypoglycaemia.

2 · Evaluation of an adrenal mass – principles

KEY POINT

Two questions: **(1) Is it malignant?** – imaging. **(2) Is it hormonally active?** – biochemistry. Either = treat (usually resect). Imaging cannot tell functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for RCC	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · Imaging

- P2 P4 Algorithm for management of an incidentaloma; Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *APW/RPW formulas, size–malignancy risk, ESE–ENSAT follow-up*
- **Unenhanced CT**: **≤10 Hounsfield units (HU)** = lipid-rich adenoma (no known malignancy); >70% of adenomas. ≤0 HU 100% specific but insensitive.
- Up to **7%** of metastases are <10 HU → biochemistry for every lesion.
- **Chemical-shift MRI** drop-out ≈ unenhanced CT.
- **CT washout** (gold standard for lipid-poor lesions, ≈30% of adenomas; phases 0, 60 s, 15 min): adenoma if APW **>60%**, RPW **>40%**. RCC/hepatoma metastases can wash out.

ALGORITHM

Mass → unenhanced CT ≤10 HU or MRI drop-out → adenoma | >10 HU → 15-min washout CT → APW >60% / RPW >40% = adenoma; below = adenoma very unlikely → manage as indeterminate

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% ACC
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** = consensus threshold (sensitive, poorly specific); 6 cm more specific. CT underestimates pathological size.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series. Age modifies.
- Suspicious: heterogeneity, irregular margins, enhancing rim, invasion. **Macroscopic fat** (<−30 HU) = myelolipoma. **Calcification is not reassuring** (up to 30% of ACC).
- Classic follow-up: re-image 6, 12 (± 24) months; ≥1 cm growth → consider resection.

Functional imaging

- P2 MIBG scan — *isotopes; best in MEN-2; cold in SDHB/VHL*
- **FDG PET**: malignancy (ACC, metastasis) when CT inconclusive; staging PPGL – best for metastases, SDHB, VHL. Poor in MEN-2 and for RCC/hepatoma metastases.
- Phaeochromocytoma >5 cm: MIBG or FDG PET pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · Hormonal work-up (all lesions >1 cm)

KEY POINT

Every incidentaloma: **1-mg dexamethasone suppression test (DST) + metanephrines**; add **aldosterone-to-renin ratio (ARR)** if hypertension or unexplained hypokalaemia. >10% are functional. **Never biopsy, sample or excise before excluding phaeochromocytoma**. CUA allows omitting metanephrines in asymptomatic <10 HU lesions.

Incidentalomas: cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5%, metastasis 2.5% → **≈18%** potentially surgical.

Cortisol

- P2 Cushing syndrome and the dexamethasone suppression test — *DST cut-offs and pitfalls, MACS*
- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes; 1.8–5.0 possible autonomy; **>5.0 µg/dL (140 nmol/L)** >95% specific. The 5.0 cut-off misses up to 18% of Cushing syndrome.
- UFC insensitive for mild excess, unreliable if GFR <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× ULN)	—

- **False positives**: paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness, seated sampling.
- Equivocal → repeat supine off β-blockers, chromogranin A (less specific if GFR <80 mL/min), or **clonidine suppression**.

5 · Biopsy and incidentaloma management

- Biopsy answers only "is this a metastasis?" (cytology cannot separate adenoma from carcinoma). ESE–ENSAT: **no biopsy without a history of malignancy**. Complications 3–12%; seeding in ACC.

> **ALGORITHM**
> Imaging characterises it? → no biopsy | functional? → no biopsy | ACC suspected? → no biopsy | will result change management (e.g., known primary) and justify risk? → biopsy, **after excluding phaeochromocytoma**

- Incidentaloma ≈5% of scans. Adenoma >85%; ≈93% inactive → no treatment; repeat hormones none (ESE) vs annual (CUA, AACE); new excess <1–1.7%.

6 · Primary aldosteronism (PA)

- P2 How will you evaluate and manage 38 yr male with uncontrolled hypertension — *ARR → saline → CT → AVS; cure ≈94%*
- Commonest secondary hypertension: 3–13% of hypertensives. Renin suppressed (vs high in secondary).
- No hypernatraemia or oedema – **mineralocorticoid escape**. Most (63–91%) **normokalaemic**.
- vs essential hypertension: stroke ≈4×, MI ≈6.5×, AF ≈12×.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – MRA
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable): AD chimeric CYP11B1 promoter + CYP11B2 coding → aldosterone under ACTH; early severe hypertension, young stroke; glucocorticoid ± MRA. **FH-III**: KCNJ5; very young, bilateral hyperplasia, severe hypokalaemia.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs incl. diuretic) or controlled on ≥4; BP ≥150/100 on three days; hypertension + hypokalaemia; **incidentaloma + hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR**: morning, seated, PAC (ng/dL) ÷ PRA (ng/mL/h). Positive: **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2. Correct K^+.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or CCB
β-blockers	Renin ↓ → **false positive**	May continue (or stop)
ACE inhibitors / ARBs	Renin ↑ → false negative	Continue; undetectable renin is suspicious
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg	PAC stays >15 ng/dL – when salt load contraindicated

- **Skip confirmation** if spontaneous hypokalaemia + undetectable renin + PAC >20 ng/dL. Only 50–87% of positive screens confirm.
- CT for all surgical candidates (excludes ACC), but ≈20% of aldosteronomas are <1 cm → CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well): K^+ corrected, morning, both adrenal veins + IVC, aldosterone and cortisol. **Selectivity**: adrenal vein/IVC cortisol >1.1–5:1 (higher with cosyntropin). **Lateralisation**: cortisol-corrected A/C ratio **>2–4:1** → ipsilateral adrenalectomy.
- **Omit AVS**: age <40 with clear unilateral adenoma >1 cm and normal contralateral gland; suspected ACC. Genetic testing if onset <20 y or family history. AVS fails: NP59, posture test.

ALGORITHM

Indication → ARR + → confirm → (family history/onset <20 → FH-I test) → surgical candidate? No → MRA | Yes → CT → age <40 + unilateral >1 cm nodule → adrenalectomy; otherwise → AVS → lateralised → laparoscopic adrenalectomy; not lateralised → MRA

- Residual hypertension: older, male, obese, long duration.
- Post-op: check PAC/ARR; † stop K^+ and MRA; high-salt diet, watch **hyperkalaemia** for weeks (contralateral glomerulosa suppressed); BP settles over 6 months.
- MRA: spironolactone 25–50 mg → max 400 mg (gynaecomastia, ED); eplerenone 25 → 100 mg (selective).

7 · Cushing syndrome and cortisol excess

- P2 Cushing syndrome and the dexamethasone suppression test — *causes, IPSS, Nelson, pituitary cure*
- Ectopic ACTH also thymic, islet cell, medullary thyroid; occult in up to 1/3. Ectopic CRH <1%.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%** – test cushingoid stone formers.

ALGORITHM

Exclude exogenous steroid → screen (1-mg DST / LNSC ×2 / UFC ×2) → abnormal → endocrinologist confirms → **morning ACTH**: low → adrenal (CT) | normal/high → pituitary MRI + **inferior petrosal sinus sampling after CRH** (central:peripheral gradient = Cushing disease; none = ectopic)

- Cushing disease: macroadenoma cure <15%, relapse up to 25%; radiotherapy slow.
- **Bilateral adrenalectomy**: failed pituitary treatment, life-threatening hypercortisolism, unresectable/occult ectopic ACTH (primary resectable in ≈10%). Cost: lifelong replacement, crisis risk, **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	AIMAH	PPNAD
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half): spotty pigmentation, Sertoli cell tumours
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of larger gland has worked in AIMAH	

- **MACS**: surgery individualised (hypertension, obesity, dysglycaemia, osteopenia); progression to overt Cushing very unlikely.
- **Post-op adrenal insufficiency expected** (atrophic contralateral gland): ≈65% after MACS, ≈99.7% after overt Cushing.
- Hydrocortisone 100 mg IV day of surgery → oral taper (25–50 mg/day). Alternative: dexamethasone 0.5 mg 6-hourly → morning cortisol <50 nmol/L confirms cure → hydrocortisone. Maintenance 10–12 mg/m²/day in 2–3 doses.
- HPA recovery median 6.5 (MACS) – 11 (Cushing) months.

8 · Phaeochromocytoma and paraganglioma (PPGL)

- P2 Phaeochromocytoma: overview, pre-operative and intra-operative management — *genetics, blockade doses, targets, post-op care, lifelong metanephrines*
- Adrenal 75–90%; paraganglioma = extra-adrenal (Zuckerkandl, bladder, para-aortic; head and neck usually non-functional). ≈0.5% of hypertension.
- Rule of 10s now: ≈24% of apparently sporadic carry germline mutations; malignant ≤5% sporadic adrenal, >1/3 extra-adrenal; children up to 40% familial, >20% bilateral.

KEY POINT

The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

- Headache 60–90%; paroxysmal hypertension 30–50%.

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	MTC, hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%

Syndrome	Gene	Features	PPGL risk / malignancy
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
NF1	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All autosomal dominant. VHL: 1 – no phaeo; 2A – phaeo, no RCC; 2B – phaeo + RCC; 2C – phaeo only.

- Metanephrines even in a suspected solitary metastasis. CT mean ≈35 HU; T2 "light bulb" sign unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes post-resection hypotension; α1-selective (incl. terazosin) less so. CCBs: sole therapy in mild cases, no reflex tachycardia. **Metyrosine** 250 mg 2–3×/day ↑ to 1.5–2 g with α-blockade for large/metastatic tumours; sedation, depression, galactorrhoea, **extrapyramidal** effects.

> KEY POINT
> **Why never β before α?** β-blockade removes β2-mediated arteriolar dilatation; catecholamines then act unopposed on α1 → hypertensive crisis. Hence selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † HR 60–70/min. **Volume repletion** key.
- **MEN-2 and VHL**: **cortical-sparing partial adrenalectomy** – recurrence possible, replacement may still be needed.
- Malignancy defined **only by metastases** (bone, lung, liver, nodes). Risk: SDHB, extra-adrenal, >5 cm, dopamine-secreting. Ablation/embolisation also; CVD responses short. 5-year survival ≈50%.
- Metanephrines at ≈6 months, then annually lifelong (recurrence even >15 years later); persistent elevation → MIBG.

9 · Non-functional benign lesions

Lesion	Key features	Management
Myelolipoma	Fat + haematopoietic tissue; macroscopic fat diagnostic; calcification >25%; may exceed 10 cm	Exception to 4-cm rule; still hormonal work-up. Excise if symptomatic or large (bleeders 10–14 cm; 6–10 cm cut-offs unvalidated); embolise bleed. Not liposarcoma (infiltrative)
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%), epithelial 9%, hydatid 7%; >30 HU = haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**; no decortication
Ganglioneuroma	Young; encases vessels without narrowing	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional; no central scar	Resect; >30% malignant (unlike renal)

10 · Adrenocortical carcinoma (ACC)

- P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *syndromes, CT, Weiss, ENSAT, R0 surgery, mitotane, trials, surveillance*
- Cortisol secretion = poor prognosis.
- Mitotic rate grades ACC and best predicts outcome; use alternative criteria for paediatric and oncocytic tumours.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Overall 5-year survival 20–47%; stage IV <5%. Recurrence 60–80% after resection.
- Resect recurrences if low grade, long disease-free interval.
- † Mitotane often needs higher-than-usual glucocorticoid doses (raises binding globulins, accelerates clearance).
- **Paediatric ACC**: >90% functional (virilisation), test TP53, better prognosis. Neuroblastoma → *Ch 63*.

11 · Adrenal metastases

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC/hepatoma may wash out → growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect isolated metastasis in selected patients – NSCLC (level 1), melanoma; RCC as cytoreduction.

12 · Adrenal insufficiency

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, HIV-related infection, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, radiation, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH/POMC)	Pallor
Electrolytes	↓Na^+, ↑**K^+**, ↑creatinine; salt craving	↓Na^+ (SIADH), normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- **Crisis** mimics acute abdomen. At risk: cortisol-secreting tumour, bilateral adrenalectomy, **previous contralateral nephrectomy/adrenalectomy** (review old notes).
- Diagnose: morning cortisol + ACTH; ACTH stimulation. Hydrocortisone ½–⅔ on waking.

KEY POINT

† **Adrenal crisis**: don't wait – draw cortisol/ACTH, **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly); 0.9% saline (1 L first hour) + dextrose; treat precipitant; taper over 1–3 days. Fludrocortisone unnecessary while hydrocortisone >50 mg/day. Sick-day rules, steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · Congenital adrenal hyperplasia (CAH)

- Autosomal recessive; **21-hydroxylase deficiency >95%** → high ACTH, hyperplasia, androgen excess → *Ch 59*.
- Adenomas (not ACC); non-classic CAH may present as adult mass. Testicular rests mimic tumour – consider before orchidectomy.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + EDP vs mitotane + streptozotocin, advanced ACC	Better response and PFS; OS similar	M-EDP first-line
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

- P2 Adrenal Sx. — *indications, contraindications, preparation, approaches, steps, complications*

1 · Surgical anatomy (essentials)

- **Right**: behind bare area of liver (fused in ~10%). Vein **<1 cm**, into **posterior IVC at 45°**; variants into right hepatic or right renal vein.
- **Left** more mobile *→ Ch 145*.

2 · Indications

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; PCC; virilising/feminising tumour
Non-functional	ACC; symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm**; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT **>10 HU**; washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake ≥ liver

KEY POINT

Trap: bilateral hyperplasia causing aldosteronism is **not** surgical — hence AVS before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC ~2%; autonomous secretion up to 20%.
- 1-mg DST for all (never spot cortisol); metanephrines for all, >2× upper limit significant.
- If hypertensive: ARR >20 with aldosterone >20 ng/dL → salt loading → AVS. Up to 40% normokalaemic.
- Sex steroids if ≥4 cm or virilisation.
- Inert, ≤4 cm, ≤10 HU, homogeneous → repeat CT at 12 months. Myelolipoma (≤ −20 HU): operate only if **>7 cm**, symptomatic or atypical; still exclude PCC.

3 · Contraindications

- Low-dose aspirin may continue. Pregnancy: necessary surgery (e.g. PCC) in any trimester. Prior nephrectomy may have removed the ipsilateral adrenal — confirm before contralateral adrenalectomy.

4 · Pre-operative preparation

Pathology	Preparation	Why
PCC	α-blockade **2 weeks** (phenoxybenzamine; doxazosin, CCB, metyrosine alternatives) → β **only after α** → volume expansion	Contracted patients collapse once vein tied. ECG, echo
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland suppressed; steroid cover dropped early mortality to zero
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia → arrhythmia

5 · Approaches

- **MIS standard** since 1992 (Gagner).
- **Retroperitoneoscopic** (lateral or **prone**): avoids bowel — prior abdominal surgery or peritonitis; robot suits the small space; bilateral small tumours prone without repositioning.
- **Open**: suspected **ACC**, large, invasion of IVC/organs. Thoracoabdominal for huge tumours or IVC extension above liver.

6 · Operative steps

ALGORITHM

Right: elevate liver → Kocher → expose IVC → up from renal vein (inferior limit) → short adrenal vein → **warn anaesthesia**, clip early → superior/lateral dissection → psoas (posterior limit) → bag, intact.

ALGORITHM

Left: reflect colon at Toldt, outside Gerota → spleen/pancreas up → follow left renal (or gonadal) vein to adrenal vein → clip after warning → protect pancreas and splenic artery → divide ganglion nexus superomedially → bag.

- **Retroperitoneoscopic**: 1-cm incision below 12th rib tip → balloon → two ports 8 cm apart. Along psoas the **renal artery is met first**. Left vein anterocephalad to renal artery; right vein posterolateral IVC above hilum — avulsion → major bleed.

KEY POINT

Ligate the vein early in PCC to limit catecholamine surge; hypotension follows — tell anaesthesia first.

7 · Special pathologies

- **PCC intra-op**: surge → nitroprusside, nitroglycerin, phentolamine; arrhythmia → esmolol.
- **ACC**: high-volume surgeon (>6, ideally ≥20/year); **open, intact, en bloc**.
- **Metastasis**: lung and kidney ≈ two-thirds. Operate when adrenal is the only disease; renal origin, <5 cm do best.
- **CAH**: bilateral adrenalectomy avoids iatrogenic Cushing; risks crisis, testicular rest tumours.
- **Ablation** (radiofrequency, >50–60 °C): avoid for PCC or ACC outside trials.

8 · Partial (cortical-sparing) adrenalectomy

- **Indications**: bilateral/hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma, cysts.
- Arteries enter superiorly, medially, inferiorly — **lateral dissection safe**; main vein may be divided (collateral outflow).
- Margins by intra-op ultrasound; indocyanine green for remnant viability. †Retain about one-third of a gland.

9 · Complications

- Bleeding: pressure, raise pneumoperitoneum, extra port, side-bite clamp IVC, convert if uncontrolled. Residual phenoxybenzamine prolongs post-PCC hypotension (volume ± α-agonist).

- †Hourly glucose 24–48 h after PCC; insufficiency → IV hydrocortisone 100 mg, saline; post-aldosteronoma hyperkalaemia → stop K^+/spironolactone, occasionally fludrocortisone.
- Flank bulge after open: protect subcostal neurovascular bundle.

10 · Outcomes

- PCC: lifelong recurrence up to **16%** → annual BP and metanephrines.
- †Aldosteronoma: hypokalaemia corrected in nearly all; hypertension cured in about half.
- Cushing: insulin resistance may persist. ACC advanced-stage 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · Anatomy of the retroperitoneum

- P1 Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. — *boundaries, contents, spaces, vessels, nerves at risk*
- Right: duodenum, IVC, posterior diaphragm, right liver, right colonic mesentery (lost fat planes = involvement). Left: aorta, splenic hilum, pancreatic tail, left colonic mesentery.
- **Depth of psoas invasion predicts nerve injury**: genitofemoral → thigh paraesthesia; femoral → weak hip flexion/knee extension (worse). Book specialist help in advance.

> **KEY POINT**
> The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · Classification

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (GCT); RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, UPS, synovial, Ewing, DSRCT, unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, SFT, PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, MPNST, ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · Clinical evaluation

- Painless/incidental; late mass effect (leg oedema, ureteric/bowel obstruction, nerve pain). Sequence: **imaging → serology → tissue**.
- **CT chest/abdomen/pelvis** first (NCCN). **Negative HU (fat)** ≈ liposarcoma, angiomyolipoma or myelolipoma. Cannot tell origin or malignancy. **MRI**: vascular involvement, contrast allergy/renal impairment. **PET-CT** selective: post-chemo seminoma residue, liposarcoma grade, lymphoma.
- Young man → **AFP, hCG** (± scrotal US); older → PSA. Blood count, electrophoresis (M-spike); † metanephrines if paraganglioma possible.
- **Biopsy** when it **changes first treatment**: lymphoma, GCT, Ewing, high-grade sarcoma for neoadjuvant therapy. **Core (14–16 G coaxial) over FNA**; † retroperitoneal (posterior) route,

tract within future resection field.

- Avoid biopsy: purely cystic lesions; † unblocked paraganglioma; raised AFP/hCG in young man. Liposarcoma core often non-diagnostic.

4 · Mesenchymal tumours (sarcoma)

- Soft-tissue sarcoma <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
WD/DD liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: DD = higher-grade non-lipogenic component; **dense nodules in fatty mass = de-differentiation** (biopsy target). WD: local recurrence up to 40%, rarely metastasises; DD: distant up to 20%, 5-year mortality ~30%. Mean time to recurrence ~**2 years** → lifelong surveillance.
- Unresectable: anthracycline (modest); trabectedin (PFS, not OS); pembrolizumab for high TMB. **Pleomorphic liposarcoma**: MDM2-negative, mimics carcinoma, early lung metastasis.
- **Leiomyosarcoma**: obstructs veins, urine, bowel; leiomyoma → STUMP → leiomyosarcoma. Biopsy underestimates aggressiveness; frequent recurrence.

> **KEY POINT**
> MDM2 amplified (WD/DD) = local disease won or lost at the margin; absent (pleomorphic) = systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy — no margin achievable.

- † **FNCLCC grade** = differentiation (1–3) + mitoses per 10 HPF (1: 0–9, 2: 10–19, 3: ≥20) + necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology (not size) predict distant spread.

5 · Neurogenic tumours

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy ± surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† → *Ch 145*

6 · Germ cell, haematological, cystic, unknown primary

- **GCT**: RP mass in a young man is GCT until proven otherwise. Extragonadal (3–5%) often a **burnt-out testicular primary**. Platinum first → resect all nonseminomatous residual masses (teratoma chemoresistant, marker-silent). Marker: i(12p). *→ Ch 79–80*
- **Sex cord stromal**: ~10% metastasise; chemo/radioresistant → early RPLND in high-risk node-negative primaries (age, >4–5 cm, necrosis, mitoses, atypia, LVI).
- **Lymphoma**: confluent nodal masses; biopsy, never resect. **Plasmacytoma**: M-spike, CD138.
- **Cystic lymphangioma** (male children, crosses midline), mucinous cystadenoma, cystic mesothelioma (women) → **complete excision** (aspiration/marsupialisation recur).
- **Unknown primary**: found in <20%. Midline/i(12p)+ → platinum; single deposit → excision/RT.

> **KEY POINT**
> Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · Surgery for RP sarcoma

- **Complete en bloc (R0) resection at the first operation** in an experienced centre; † the first operation is the best chance.
- **Compartmental resection**: remove adherent uninvolved neighbours (kidney, colon, psoas fascia) → less local recurrence. Preserve duodenum, pancreas head, major vessels, femoral nerve unless invaded. † **Never enucleate** (pseudocapsule contains tumour).
- Nephrectomy in up to **55%**; rarely dialysis — † split renogram first. ~5% lose femoral/obturator nerve → knee brace.
- CT within 6 weeks; no bowel prep. GCT: markers within 7–10 days, sperm banking; prior **bleomycin** → low FiO_2, restrictive fluids.
- Complete IVC thrombosis simplifies surgery (collaterals; resect); partial may need PTFE/allograft.
- Midline laparotomy; mesenteric root/Toldt mobilisation. **Split-and-roll** over great vessels; ligate lumbar vessels and retrocrural lymphatics (chylous ascites); protect sympathetic trunks, genitofemoral nerve.

- **Aorta always reconstructed**. **IVC**: chronically occluded → ligation tolerated; grafts thrombose 10–40% → anticoagulate ≥3 months.

8 · RPLND in germ cell tumours

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *landing zones, templates, PC-RPLND, nerve-sparing*
- Primary RPLND survival ~100% with either template. PC-RPLND: full bilateral infrahilar standard; unilateral only if **≤5 cm** in landing zone, normal markers, IGCCCG good/intermediate risk.
- Nerve-sparing (L1–L4 fibres, superior hypogastric plexus below IMA) → antegrade ejaculation **90–100%**; temporary anejaculation = neuropraxia.

9 · Complications

- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *risk, diagnosis, stepwise treatment*
- Chylous ascites 2–7% post-chemotherapy (suprahilar dissection, IVC resection). Pancreatic leak after left suprahilar work → drain amylase. VTE ~1 in 5 after caval reconstruction. Femoral neuropraxia from retractors; paraplegia after combined mediastinal + RP resection.

ALGORITHM

Chylous ascites: paracentesis + MCT diet ± octreotide → drain → NBM + TPN → >100 mL/day: peritoneovenous (LeVeen) shunt or ligation → MCT diet 1–3 months after resolution.

10 · Adjuvant and neoadjuvant therapy

- **RT**: pre-op better tolerated than post-op (50–60 Gy bowel toxicity). **STRASS**: no gain → **not standard**; † exploratory benefit in (low-grade) liposarcoma keeps it an option. IORT/brachytherapy investigational.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma largely resistant; myxoid liposarcoma, synovial, Ewing respond — **standard neoadjuvant only for Ewing**. Doubtful resectability → induction multimodal.

11 · Recurrence and follow-up

- † Local recurrence kills in liposarcoma; leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually — lifelong for liposarcoma.
- Re-resect if resectable and slow (WD liposarcoma); each later R0 less likely.

12 · Landmark trials

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal RFS difference (HR 1.01); more complications	Pre-op RT not standard
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · Adrenal incidentaloma: two questions

ALGORITHM

1. Malignant? Unenhanced CT ≤10 HU or MRI drop-out → adenoma. >10 HU → washout CT (APW >60% / RPW >40% = adenoma). >4 cm, suspicious or growth >1 cm → consider resection (→ *Ch 146*).

2. Functioning? All: 1-mg DST, metanephrines; ARR if hypertensive/hypokalaemic.

Silent, <4 cm, benign → no follow-up.

B · Functional tumours compared

	Aldosteronism	Cushing	Phaeo
Screen	ARR	1-mg DST, LNSC, UFC	Metanephrines
Next	Confirm → CT → AVS	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics for all
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers (months)	Hypotension, hypoglycaemia
Medical	MRA (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · Adrenalectomy approach

KEY POINT

Laparoscopic (transperitoneal or retroperitoneoscopic) for benign tumours. Open for large suspected ACC or invasion (rupture → spill, recurrence). Right: short vein into IVC — control carefully. Partial adrenalectomy for bilateral/hereditary disease (→ *Ch 147*).

D · Retroperitoneal mass

ALGORITHM

Image → exclude paraganglioma (metanephrines), GCT (testis, markers), lymphoma before biopsy.

Core biopsy, retroperitoneal route, if it changes treatment.

Sarcoma: en bloc R0 at first operation (multivisceral if needed), specialist centre; never enucleate. Lifelong CT, especially liposarcoma (→ *Ch 140*).